

AQUADILLA HERNANDEZ INTL, Puerto Rico

WMO: 785140

Lat: 18.498N

Lon: 67.129W

Elev: 67

StdP: 100.53

Time Zone: -4.00 (W04)

Period: 94-17

WBAN: 11603

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.8	20.1	15.2	10.9	22.6	16.4	11.8	22.6	12.8	27.3	11.1	26.3	2.3	120	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	32.2	25.9	31.8	25.9	31.1	25.9	27.0	30.3	26.6	30.0	26.2	29.7	6.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.1	21.7	29.7	25.6	21.0	29.5	25.2	20.4	29.2	85.4	30.4	83.5	30.2	82.2	30.0	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	10.2	9.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.0	24.4	24.2	24.5	25.4	26.1	26.8	27.2	27.6	27.3	27.0	26.1	25.1
	DBStd	1.49	1.07	0.87	0.85	0.89	0.91	0.74	0.88	0.95	0.83	0.85	1.04	0.99
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5832	446	398	448	463	498	505	534	545	519	526	483	467
	CDD18.3	2791	187	164	190	213	239	255	276	287	269	268	233	209
	CDH23.3	22834	1046	943	1202	1597	1934	2367	2724	2883	2601	2363	1814	1359
(13)	CDH26.7	6174	165	148	210	355	480	672	848	979	842	729	470	276
(14)	Wind	WSAvg	4.5	5.0	5.0	5.0	4.3	4.5	5.0	4.5	3.6	3.4	4.1	4.6
(15) Precipitation	PrecAvg	1561	70	79	93	139	192	168	115	145	148	166	152	120
	PrecMax	2112	144	191	286	383	499	342	250	266	283	376	386	272
	PrecMin	1067	3	4	17	17	47	57	39	37	53	50	42	22
	PrecStd	271	37	51	62	104	101	68	53	62	68	79	89	66
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	31.0	30.0	30.9	31.2	31.7	32.2	33.2	32.9	33.0	32.9	31.8
		MCWB	21.9	24.5	24.4	24.8	25.5	26.3	25.7	26.4	25.9	26.0	26.9	22.7
	2%	DB	29.1	29.0	29.1	30.0	30.4	30.9	31.8	32.1	31.8	31.2	31.0	30.0
		MCWB	24.3	23.6	23.9	24.6	25.3	25.8	25.6	26.1	26.1	26.3	26.0	25.4
	5%	DB	28.2	28.1	28.6	29.2	29.9	30.2	31.0	31.2	31.0	30.5	30.0	29.0
		MCWB	23.7	23.4	23.7	24.0	25.2	25.7	25.6	26.1	26.0	26.0	25.5	24.5
	10%	DB	27.4	27.5	28.0	28.9	29.1	29.9	30.1	30.8	30.2	30.0	29.1	28.2
		MCWB	23.2	23.1	23.4	23.8	24.8	25.5	25.6	26.0	25.7	25.7	25.0	23.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.8	25.7	25.9	26.0	26.6	27.0	27.2	27.5	27.4	27.3	26.5	26.1
		MCDB	28.3	27.9	28.2	29.4	29.8	30.1	30.1	30.6	30.7	30.2	29.8	28.9
	2%	WB	25.0	24.9	25.1	25.2	26.0	26.2	26.3	26.9	26.5	26.7	26.1	25.4
		MCDB	27.8	27.6	27.9	28.9	29.4	29.6	29.9	30.5	30.3	29.9	29.4	28.6
	5%	WB	24.1	24.1	24.6	24.8	25.6	26.0	26.1	26.3	26.1	26.2	25.7	24.8
		MCDB	27.2	27.1	27.7	28.5	29.0	29.5	29.8	30.0	30.0	29.6	29.0	28.2
	10%	WB	23.5	23.6	23.8	24.2	25.1	25.6	25.8	26.0	25.9	25.9	25.2	24.2
		MCDB	26.6	26.8	27.3	28.0	28.5	29.1	29.5	29.7	29.7	29.3	28.5	27.5
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.1	7.3	6.8	6.5	6.4	6.4	6.8	6.6	6.9	6.6	6.9
		MCDBR	7.9	8.1	8.0	7.5	7.4	6.8	7.3	7.5	7.4	7.6	7.7	7.8
		MCWBR	4.7	4.8	4.8	4.1	4.2	4.0	3.9	3.8	4.0	4.3	4.0	4.6
	5% WB	MCDBR	7.0	7.6	7.7	7.0	6.8	6.5	6.4	6.7	6.5	6.6	6.5	6.9
		MCWBR	5.0	5.6	5.7	4.5	4.4	4.2	4.0	4.2	4.0	4.3	4.0	4.8
(40) Clear-Sky Solar Irradiance	taub		0.365	0.384	0.382	0.412	0.450	0.524	0.518	0.485	0.435	0.386	0.383	0.374
	taud		2.574	2.510	2.514	2.449	2.363	2.112	2.128	2.252	2.428	2.578	2.585	2.580
	Ebn at Noon		920	917	927	895	851	782	788	821	865	904	895	899
	Edh at Noon		95	106	108	117	126	160	158	140	117	97	93	92
(44) All-Sky Solar Radiation	RadAvg		4.62	5.37	5.97	6.27	6.04	6.12	6.20	6.08	5.72	5.18	4.51	4.36
	RadStd		0.18	0.11	0.24	0.33	0.40	0.30	0.27	0.19	0.40	0.33	0.34	0.20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.33	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+112

Nomenclature: See separate page